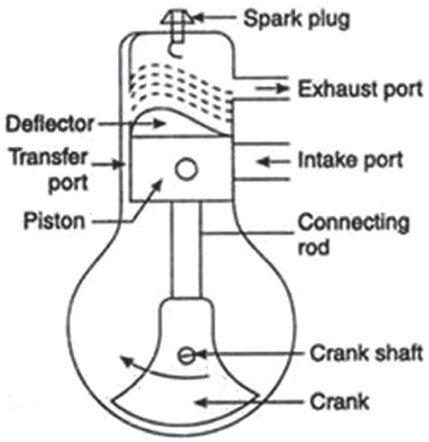


Figure 5.5(a) Two Stroke Petrol Engine

Working principle

In two stroke engine, all the four events namely suction, compression, power and exhaust cannot be distinctly identified. For each revolution of crankshaft or for every two stroke of piston, a working / power stroke is obtained. Hence the cycle of operation can be explained with two stroke of piston movement namely Upward stroke and Downward stroke.

Upward Stroke

During upward stroke, the piston moves from BDC to TDC. It expels the burnt gases to the atmosphere through the exhaust port. It closes the transfer port and then the exhaust port. Then it compresses the already inducted charge (air-fuel mixture) in the combustion chamber of the cylinder. At the end of the upward stroke, the ignition of the fresh charge is takes place by the spark plug.

Further, the upward movement of the piston a partial vacuum is created in the crankcase and this allows the entry of the fresh charge into the crankcase through uncovered inlet port. The exhaust port and the inlet port remains covered when the piston at the TDC.

Downward stroke:

As soon as the combustion of the fresh charge takes place, a large amount of the hot

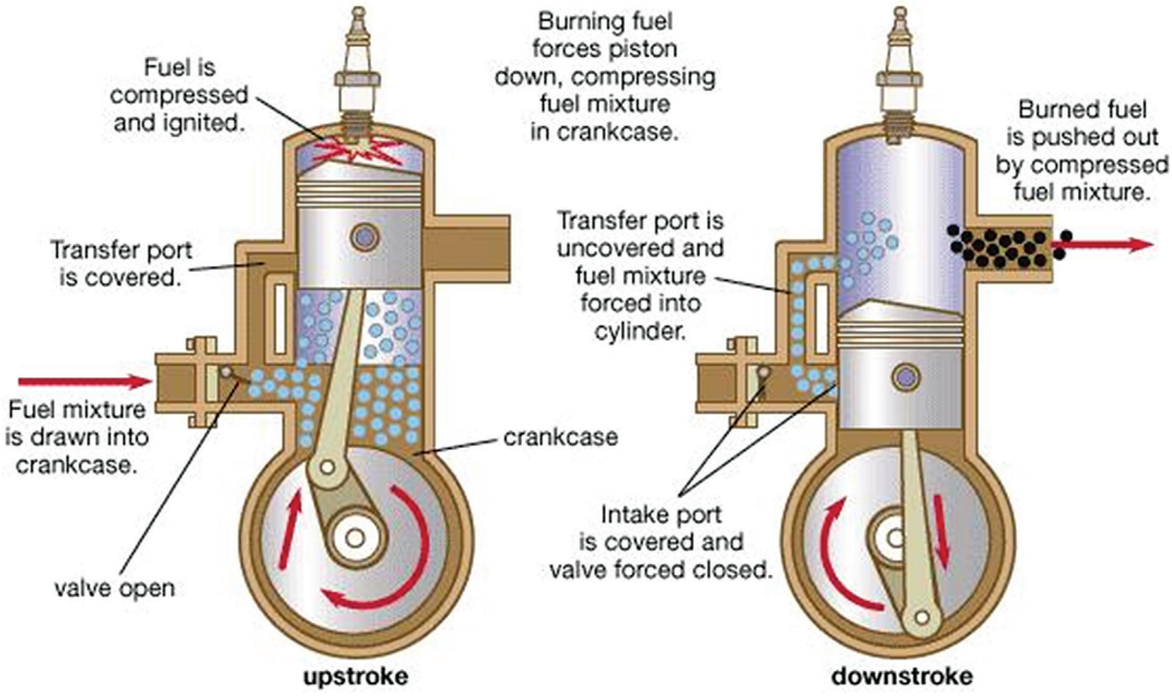


Figure 5.5(b) Upword and Downword Stroke of Two Stroke Petrol Engine

AM_Unit_05.indd 102 12/20/2019 5:26:39 PM